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Elastic moduli of normal and pathological human breast tissues: an inversion-technique-based investigation of 169 samples

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Abstract

Understanding and quantifying the mechanical properties of breast tissues has been a subject of interest for the past two decades. This has been motivated in part by interest in modelling soft tissue response for surgery planning and virtual-reality-based surgical training. Interpreting elastography images for diagnostic purposes also requires a sound understanding of normal and pathological tissue mechanical properties. Reliable data on tissue elastic properties are very limited and those which are available tend to be inconsistent, in part as a result of measurement methodology. We have developed specialized techniques to measure tissue elasticity of breast normal tissues and tumour specimens and applied them to 169 fresh *ex vivo* breast tissue samples including fat and fibroglandular tissue as well as a range of benign and malignant breast tumour types. Results show that, under small deformation conditions, the elastic modulus of normal breast fat and fibroglandular tissues are similar while fibroadenomas were approximately twice the stiffness. Fibrocystic disease and malignant tumours exhibited a 3–6-fold increased stiffness with high-grade invasive ductal carcinoma exhibiting up to a 13-fold increase in stiffness compared to fibroglandular tissue. A statistical analysis showed that differences between the elastic modulus of the majority of those tissues were statistically significant. Implications for the specificity advantages of elastography are reviewed.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

The elastic properties of breast tissues have been of increasing interest for many medical applications. Elastography is an imaging technique developed over the past two decades for imaging soft tissue elastic modulus (Ophir *et al* 1991, Muthupilai *et al* 1995) that may offer new insights into tissue diagnosis. This technique was developed as a more quantitative alternative to manual palpation that is commonly used as a means to detect breast cancer. Unlike palpation, which is essentially qualitative and only useful to detect relatively large and superficial tumours, breast elastography is capable of detecting smaller, deeper tumours and providing quantitative information on their stiffness as characterized by elastic modulus (Garra *et al* 1997, Samani *et al* 2001b, McKnight *et al* 2002, van Houten *et al* 2003, Sinkus *et al* 2005). There are two classes of elastography techniques: quasi-static elastography and harmonic elastography. While harmonic elastography techniques are developed for tissue elastic modulus reconstruction, in quasi-static ultrasound elastography modulus reconstruction is usually not done and only a strain image is considered to approximate the modulus distribution. The elastic moduli of soft tissues are known to depend on the molecular makeup of tissue components as well as the macroscopic features of tissue structure involving stromal tissue distribution and connective elements (Fung 1993). Gross pathological assessment and histological analysis confirm that tissue alterations such as breast cancer lead to changes in macroscopic and microscopic structural reorganization.

Modelling of breast tissue mechanics is another subject of interest with numerous applications such as medical imaging and computer-assisted interventional procedures. To improve the outcome of breast needle biopsy Azar *et al* (2001) developed a finite element (FE) model for the breast to predict tissue deformation during the procedure. In Samani and Plewes (2002), we presented an FE model we developed to calculate breast tissue deformation resulting from its own weight. The objective of that study was to update data of breast MRI images taken in prone position to more accurately reflect the supine breast during surgery. Over the past few years, non-rigid registration techniques based on tissue biomechanics have been developed. Samani *et al* (2001a) proposed an FE-based approach for breast non-rigid registration while Schnabel *et al* (2003) used an FE breast model to validate a free-form deformation non-rigid registration technique. More recently, Carter *et al* (2005) presented an FE model that can be potentially applied for breast image guided surgery while Chung *et al* (2005) proposed a contact mechanics model to simulate tissue deformation of breast undergoing x-ray mammography. For these applications, computational methods such as the FE method have been developed and are well known to simulate tissue deformation with reasonable accuracy. However, the ability of these FE models to predict *in vivo* behaviour strongly depends on the accuracy of the assumed mechanical properties of tissue components.

Several investigations have been conducted to characterize the biomechanical properties of soft tissues that are subject to some degree of mechanical activity such as muscles and arteries (Fung 1993, Humphrey 1995). However, very limited quantitative information is available on the biomechanical properties of soft tissues, which do not have an active mechanical function such as the breast, liver, brain or prostate tissues. One key study by Sarvazyan *et al* (1994) presented results in which they measured the elastic modulus of 168 *ex vivo* specimens of normal, fibroadenomatous and cancerous breast tissues. This indicated that the range of variations of Young's modulus of breast tissues to be approximately two orders of magnitude extending from a fraction of kPa for normal tissues to over 20 kPa for carcinomas with micro-calcifications. Krouskop *et al* (1998) measured the elastic modulus of 142 *ex vivo* samples of normal and pathological breast tissues and concluded that the Young's modulus of breast tissues is highly dependent on the level of tissue precompression used in the measurement.

For example, the Young's modulus of invasive carcinoma was found to be 5 and 25 times larger than that of normal adipose tissue when applying precompression levels of 5% and 20%, respectively. This dependence confirms the nonlinear elastic behaviour often observed in biological tissues. More recently, a number of investigators have measured breast tissue elastic modulus *in vivo* based on magnetic resonance elastography (MRE) (Kruse *et al* 2000, van Houten *et al* 2003, Sinkus *et al* 2005). The results of some of these studies have been summarized by Van Houten *et al* (2003) which clearly indicate a wide variation in modulus not only among different types of tissues but also within each tissue type and was most evident in normal fat and fibroglandular tissues.

The purpose of this paper is to catalogue the results of elastic modulus measurements of normal and pathological *ex vivo* human breast tissues based on two measurement methods. For normal tissues, we used a technique described in Samani *et al* (2003) where small block shape specimens were indented and the resulting force–displacement slope was converted to the tissue's Young's modulus using an FE model. For tumours, we used another measurement technique we described in Samani and Plewes (2007). In that technique, tumours within tissue slices, composed of the tumour surrounded by normal tissues, were indented and the resulting force–displacement slope converted to the tumour's Young's modulus iteratively using the tissue slice FE model. We used these techniques to measure the Young's modulus of 169 normal and various pathological breast tissue specimens and conducted a statistical analysis to determine which tissue would be distinguishable on the basis of their elastic modulus alone.

2. Materials and methods

To measure the tissue elastic modulus, we used an indentation technique in which a tissue specimen's mechanical response is characterized by its force–displacement slope as it was undergoing indentation by a small flat-ended cylindrical indenter. We developed two systems for measuring the Young's modulus of breast tissues, one for normal tissues and the other for tumours. These systems have been described in detail in earlier work (Samani *et al* 2003, Samani and Plewes 2007); however, in the next two sections, these methods will be briefly reviewed.

2.1. Elastic modulus measurement of normal breast tissues

In order to measure the Young's modulus of normal breast tissue including fat and fibroglandular tissues, we developed a technique, which makes the assumption that a uniform specimen of tissue can be obtained. It allows a repeatable measurement of Young's modulus from a small tissue specimen without having a negative impact on the proceeding pathological assessment. This technique can be applied to normal tissue samples in addition to large tumours from which a homogeneous tissue sample can be excised while keeping the tumour margin intact. In this measurement system, $\sim 15 \times 15 \times 10 \text{ mm}^3$ block shape tissue samples were dissected from post-surgical breast tissues, usually obtained from reduction mammoplasties. These blocks were transported from the surgery to our elasticity measurement lab within 2 h post-resection for analysis. Similar to Krouskop *et al* (1998), we have not observed measurable changes in the data obtained after allowing the specimens to sit for periods up to 2 h. A block shape tissue specimen was obtained using a specialized cutting technique illustrated in the referred earlier publications. After cutting, the tissue block was inspected for any obvious indications of inhomogeneity or macroscopic tissue tears that may render the specimen inappropriate for elasticity measurement. Using the computer-controlled loading

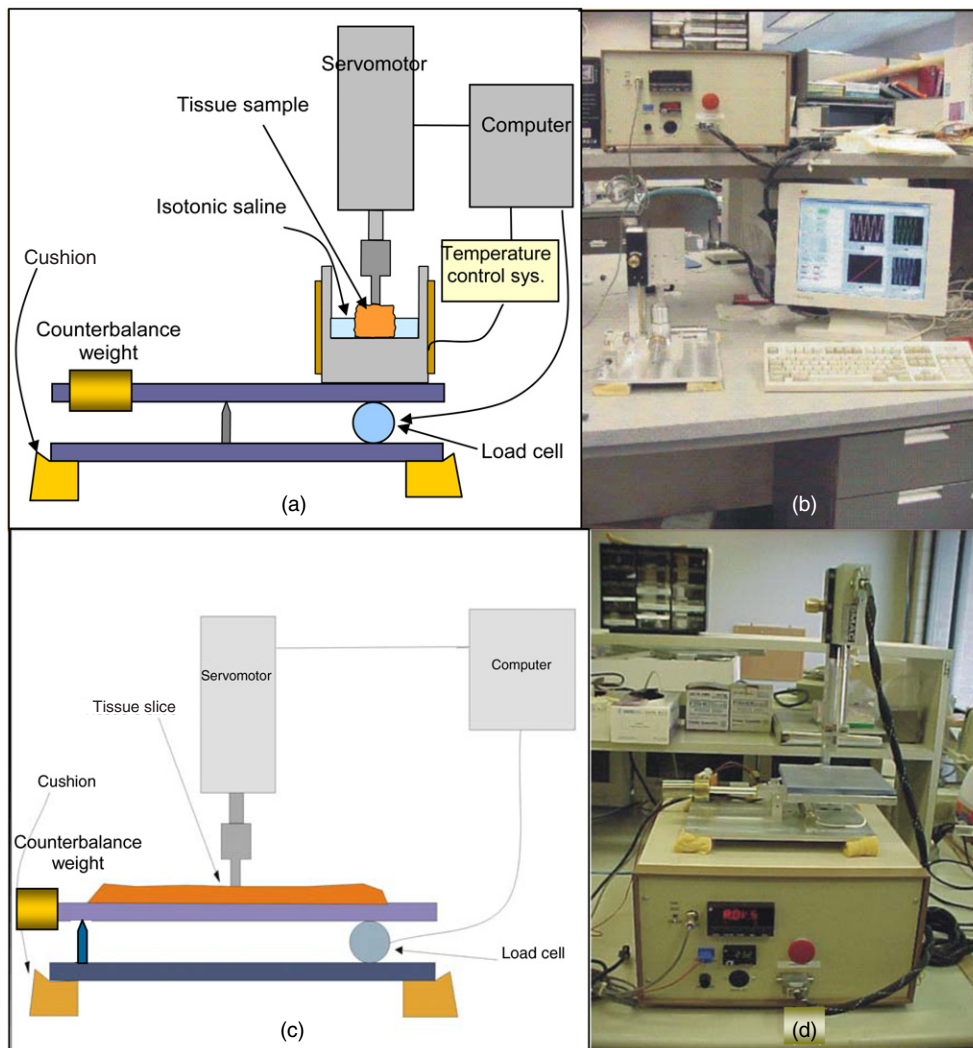


Figure 1. Breast tissue modulus measurement system. (a) Schematic of the block shape tissue indentation apparatus, (b) photo of the block shape tissue indentation apparatus, (c) schematic of the tissue slice indentation apparatus and (d) photo of the tissue slice indentation apparatus.

system (figures 1(a) and (b)), each tissue specimen was indented with a small flat-ended cylindrical indenter with 5 mm diameter. Throughout the test, the tissue specimen was placed in a short cylindrical sample holder while it was surrounded by a thin layer of saline to maintain its moisture. In our previous study, we concluded that using a temperature control system to maintain the sample's temperature at the biological 37 °C did not have a measurable effect on the measured Young's modulus. Thus, the tissue specimens were tested in room temperature. The indenter was driven by a servo-controlled system with a controller programmed to apply a sinusoidal motion with peak-to-peak amplitude of 0.5 mm and a frequency of 0.1 Hz. A load cell was used to measure forces resulting from tissue indentation. The forces were recorded along with corresponding indenter positions for future offline processing. Before acquiring the force and displacement data, 25 cycles of motion were applied to the tissue for

preconditioning (Fung 1993). After preconditioning, the number of cycles was set to five for which force–displacement data were acquired. To improve the accuracy, the measurement was repeated four times with different sample rotations allowing 3–5 min prior to the next test to allow the tissue to relax between experiments. Following this procedure, the tissue specimens were placed in a container, submerged by formalin and sent to a pathology lab for histological assessment.

To obtain the tissue's Young's modulus E , we used the following equation:

$$E = \kappa S. \quad (1)$$

In this equation, κ is the conversion factor, which was calculated using FE modelling and S is the measured force–displacement slope. The FE mesh was created using the actual geometry of each block shape tissue specimen obtained from measuring the dimensions of each block. As described in Samani *et al* (2003), the tissue was idealized as a homogeneous, linear elastic and incompressible material. Thus, the FE model simulated the indentation test of each tissue specimen with an arbitrary Young's modulus and a Poisson's ratio of $= 0.495$ (Fung 1993). To account for tissue incompressibility, the FE model was developed by adding a mathematical constraint using a Lagrange multiplier. With such a constraint, the analysis results become less sensitive to uncertainties in the value of Poisson's ratio. Assuming slipping boundary conditions at the indenter's contact area and sample's bottom, a quasi-static loading corresponding to 0.5 mm indentation was simulated and the force–displacement slope S was obtained. According to equation (1), the conversion factor κ of the tissue sample will be the ratio of the arbitrarily input Young's modulus to the calculated slope. The obtained specimen-specific conversion factor was, then, multiplied by the measured force–displacement slope to calculate the tissue's Young's modulus. The force–displacement slope was obtained using data acquired from the second loading cycle. For that purpose, we used a least-squares fit of loading data points corresponding to displacement values smaller than 0.2 mm. As explained in our earlier publications, the measured forces of data points corresponding to the unloading portion are underestimated due to tissue viscoelasticity, in that unloading does not result in immediate tissue deformation recovery. As such, there are time windows throughout the unloading phase where the indenter is not in contact with the tissue, which renders force reading during this phase unreliable.

2.2. Elastic modulus measurement of breast tumour tissues

For tumour tissues, we used a different measurement system as recently described by our group (Samani and Plewes 2007). Unlike the system we used with normal tissues, this system does not require cutting the tumour and allows it to remain intact within breast tissue specimen. In this technique, a region of the specimen corresponding to the tumour was indented and the resulting force–displacement slope was converted to the tumour's Young's modulus using an iterative inversion technique. Compared to the indentation apparatus we used for normal tissues, the indentation apparatus used with the tumour samples was modified to accommodate slice geometry and altered measurement procedure. This apparatus is illustrated in figures 1(c) and (d).

The inversion technique used to obtain the tumour's Young's modulus is based on the conversion factor concept and employs a numerical procedure that finds the tumour's Young's modulus by iteratively improving the conversion factor. This technique is summarized in the flow chart shown in figure 2. For simulating the force–deformation slope, we used an FE model of the given breast tissue slice. The tissue slice was idealized as linear elastic piecewise homogeneous material with two distinct regions: tumour region and surrounding normal tissue

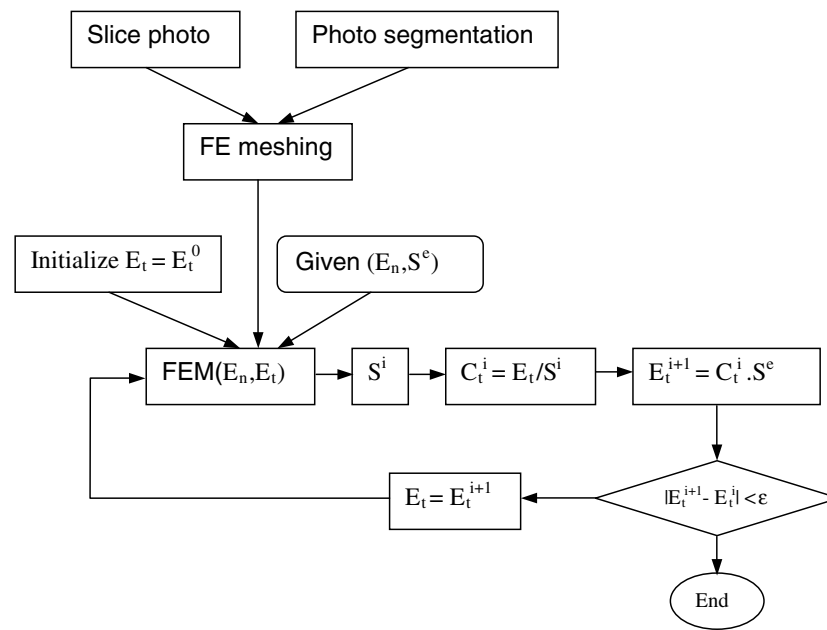


Figure 2. Flow chart of the iterative inversion algorithm used to calculate the Young's modulus of tumours within tissue slices. In this flow chart, E_t , E_t^0 and E_n are the current estimate of the tumour's Young's modulus, its initial guess value and the normal tissue's Young's modulus, respectively. S^e and S^i are the measured and calculated force–displacement slopes, respectively, and C_t^i and ε are the current estimate of the tumour's conversion factor and a small number, respectively.

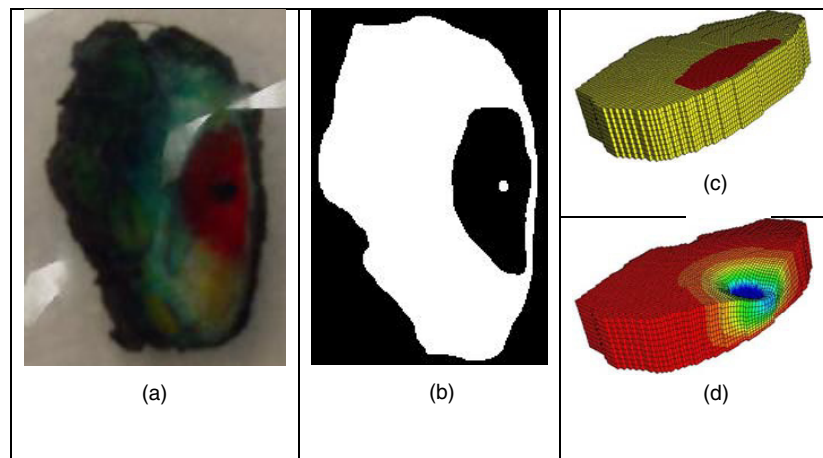


Figure 3. Illustration of the FE meshing process. (a) Photo of a small breast tissue slice obtained from a lumpectomy procedure, (b) image of the manually segmented photo in (a), (c) calculated FE mesh of the tissue slice and (d) deformed shape of the tissue slice obtained from indentation FE simulation.

region. A typical tissue slice photograph, its corresponding segmented photo and FE model before and after indentation are shown in figure 3.

To indent breast tissue, slices consisting of tumours within normal tissues were obtained from lumpectomy procedures or from slicing whole breasts acquired after mastectomy. Similar

to the normal tissue samples, the slices were transported to the elasticity measurement lab within 2 h after surgery. The tumour was first dyed and then photographed. Dying the tumour was done for two purposes: first to facilitate slice photo segmentation of tumour area necessary for creating slice FE mesh and second to identify the tissue indentation area in the FE model. After acquiring a digital photograph of the tissue, the slice height at the tumour's indentation area was measured with the loading apparatus. Each tissue specimen was indented with a small flat-ended cylindrical indenter with 5 mm diameter. Preload was set to a maximum value of 3.0 g, and as with normal tissues the tumour was loaded with 25 sinusoidal cycles for preconditioning. This was followed by five sinusoidal cycles using a peak-to-peak amplitude of 0.5 mm and frequency of 0.1 Hz. This measurement was repeated four times to improve accuracy, where each time the tissue was allowed to relax for a 3–5 min prior to the next test. To avoid tissue dehydration, a few drops of saline were applied to the slice's top surface throughout the experiment. The force–displacement data for each test were stored in a separate file for later offline processing. Similar to measurement of normal tissues, we used data acquired from the second loading cycle and performed a least-squares fit of data points corresponding to displacement values smaller than 0.2 mm. These data were then used to obtain the experimental force–displacement slope required to calculate the tumour's Young's modulus as described above.

2.3. Pathological assessment of tissue specimens

Fresh tissue was collected within 30 min of surgical resection. For tissue specimens obtained from reduction mammoplasties or from very large tumours, a specimen approximately $30 \times 30 \times 15 \text{ mm}^3$ was dissected manually to obtain a region of homogeneous appearance. The specimen was placed in a small container labelled by a code specific to the patient for pathology tracking and then transported to the elasticity measurement lab. For slices acquired from lumpectomy or mastectomy, a slice containing a tumour was selected after gross inspection of the tissue. The slice orientation was identified and the superior (S) and anterior (A) sides were properly marked on the slice. After the modulus measurement, the tissue specimens were processed into paraffin blocks following neutral buffered formalin fixation. From each of those blocks, a slide stained with haematoxylin and eosin was prepared and used for determining the tissue type or for pathological assessment of lesions using optical microscopy. The tissue histology of each specimen, including normal tissue specimens, was determined by a single pathologist (Judit Zubovits) assigned to this study, who is trained in clinical breast pathology.

3. Results

Table 1 indicates that the 169 tissue specimens were composed of 71 fat, 26 fibroglandular, 16 fibroadenomatous, 4 tissues with fibrocystic disease (FSD), 4 infiltrating lobular carcinoma (ILC), 4 ductal carcinoma *in situ* (DCIS), 12 low-grade infiltrating ductal carcinoma (IDC), 21 intermediate-grade IDC, 9 high-grade IDC, 1 invasive mucinous carcinoma (IMC) and 1 fat necrosis. With the exception of a small number of normal tissue specimens, all other specimens came from different subjects. The 97 normal fat or fibroglandular tissues were tested using the technique described in section 2.1 while the remaining abnormal tissue specimens were slices containing tumour tissues that we tested using the technique described in section 2.2.

Figure 4 shows typical graphs of the loading and unloading parts of second force versus displacement cycles corresponding to four tissue samples. Two of those samples were normal fat and glandular tissues while the other two were identified as fibroadenoma and invasive

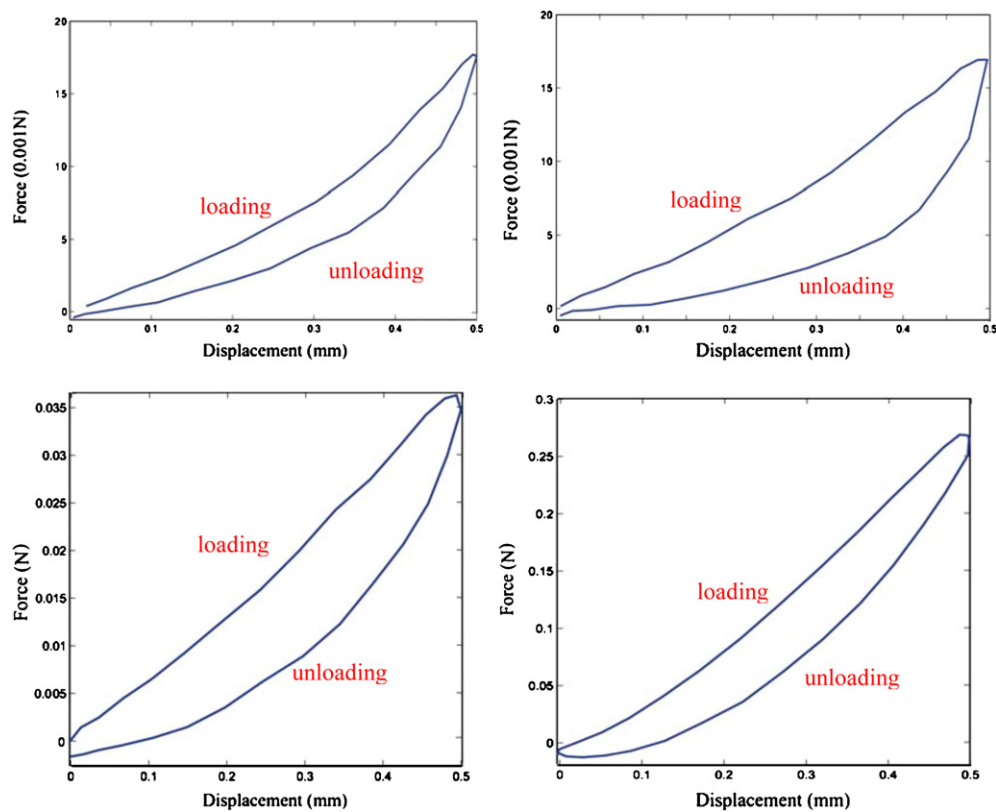


Figure 4. Typical graphs of the second force versus displacement cycles corresponding to (a) normal fat tissue, (b) normal glandular tissue, (c) fibroadenoma and (d) invasive ductal carcinoma (IDC).

Table 1. Mean and standard deviation values of the calculated Young's modulus summarizing the tested 169 breast tissue samples.

Breast tissue type	Number of samples	Young's modulus (kPa)
		mean $\pm$ STD
Normal fat	71	3.25 ± 0.91
Normal fibroglandular tissue	26	3.24 ± 0.61
Fibroadenoma	16	6.41 ± 2.86
Low-grade IDC	12	10.40 ± 2.60
ILC	4	15.62 ± 2.64
DCIS	4	16.38 ± 1.55
Fibrocystic disease	4	17.11 ± 7.35
Intermediate-grade IDC	21	19.99 ± 4.2
High-grade IDC	9	42.52 ± 12.47
IMC	1	20.21
Fat necrosis	1	4.45

ductal carcinoma. Table 1 presents the means and standard deviations of the elastic moduli of the tested samples. This shows that fat and fibroglandular tissues have similar stiffness

Table 2. p -values of data set pairs corresponding to tested tissues obtained using a Student's t -test. Shaded cells highlight pairs with p -values higher than 0.005, i.e. pairs with no statistically significant difference.

	Fat	Fibrogland	Fibroaden	IDC (low)	ILC	DCIS	Fibrocystic	IDC (inter)	IDC (high)
Fat	n.a.	0.71	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
Fibrogland.		n.a.	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
Fibroaden.			n.a.	0.0008	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
IDC (low)				n.a.	0.0037	0.0007	0.0130	<0.0001	<0.0001
ILC					n.a.	0.6400	0.7200	0.0590	0.0016
DCIS			Symm.			n.a.	0.8500	0.1100	0.0018
Fibrocystic.							n.a.	0.2800	0.0033
IDC (inter)								n.a.	<0.0001
IDC (high)									n.a.

under small strains. We found that fibroadenomas were approximately 2 times stiffer than fibroglandular tissue, while high-grade IDC tumours are 13 times stiffer than normal tissues. The remaining benign and cancerous tissues were found to be 3–6-fold stiffer than normal fibroglandular tissue.

The statistical significance of the differences in the obtained Young's moduli was assessed using a Student's t -test. To perform this assessment, data from all possible tissue type pairs were processed. Table 2 summarizes p -values obtained from each pair of data sets of the measured Young's modulus. To determine whether the difference between two sets is statistically significant, we used a threshold p -value of 0.005. Accordingly, we have determined that with the exception of pairs corresponding to shaded cells in table 2 the differences between all other pairs are statistically significant. This means that based on the data obtained in this research it is not reliable to make distinction between only certain types of tissues based on their Young's modulus. For example, while fibroadenoma and high-grade IDC can be reliably distinguished from all other normal and pathological tissues, DCIS cannot be reliably distinguished from ILC, fibrocystic disease or intermediate-grade IDC diseases. Moreover, this shows that fibrocystic disease is not differentiable from many other pathological tissues, including DCIS, ILC and low- and intermediate-grade IDC.

4. Discussion and conclusions

In our previous research, we presented two indentation measurement systems that were developed to measure the Young's modulus of breast tissue samples. This work applies these methods in the assessment of 169 normal and pathological breast tissue specimens. These data show that fat and fibroglandular tissues exhibit virtually identical mechanical properties with a Young's modulus of 3.25 kPa under small strains. This is in general agreement with the results obtained by Sarvazyan *et al* (1994) who conducted their study on *ex vivo* breast tissue specimens and Sinkus *et al* (2005) who conducted *in vivo* breast MRE study. This implies that image segmentation of fat and fibroglandular tissues may not be necessary when using an FE mesh to simulate breast tissue deformations under small strain. However, large-scale deformations of the breast for surgical simulation will likely be influenced by the tissue nonlinearity clearly illustrated in the loading data shown in figure 4.

Among the various other pathologies, most cancers exhibited a substantially higher Young's modulus than fibroglandular tissue. It is notable that fibrocystic disease also presented

with an elevated modulus, a feature consistent with its histological phenotype. These data indicated that high-grade invasive ductal carcinomas were the stiffest tumours exhibiting a Young's modulus approximately 13 fold larger than either fat or fibroglandular tissue with most other tumours demonstrating a 3–6-fold increase in tissue stiffness.

Table 1 summarizes the mean and standard deviation values of the tissue Young's moduli and indicates that the standard deviation is high in some cases, e.g. high-grade IDC. This can be attributed to a number of factors including having a small statistical sample, systematic errors associated with the measurement techniques and used FE models, tissue heterogeneity and finally to the variability of tissue stiffness during menstrual phase and for different age groups. Table 1 shows that the sample sizes of ILC, tissues with fibrocystic disease and high-grade IDC were four, four and nine, respectively, which are relatively small. Systematic errors such as those resulting from the manual outlining of the tumour boundary, segmentation of tissue components as well as ignoring thickness variations of tissue slices may have a direct impact on the accuracy of the tissue's FE model. These errors were discussed in Samani and Plewes (2007) and it was concluded that the segmentation error, especially with larger tumours, is insignificant. Heterogeneity of tumour tissues is another important factor that contributes to high Young's modulus variability. It was observed by both visual tissue inspection and pathological assessment using a microscope that tumour tissues often consist of piecewise homogeneous abnormal tissues with pockets of normal tissues randomly distributed to form the tumour. Lorenzen *et al* (2003) showed that there is a variability of up to 35% in the stiffness of fibroglandular tissues throughout the menstrual cycle. This partially justifies the variability observed in the Young's modulus of fibroglandular tissues. Using the loading part of the force–displacement curve and ignoring the unloading part is an issue that has an impact on the reliability of using data obtained from this study in interpreting elastography images. The reason for using the loading curve was two fold: first is the fact that the measured initial portion of the unloading curve is underestimated as discussed in the methods and second is the tissue idealization as an elastic material and ignoring its viscoelasticity.

By comparing our results with those reported by Sarvazyan *et al* (1994), we note that some of the results agree well while the other ones show Young's modulus values that are generally smaller than ours. For example, while there is a general agreement between the Young's modulus values reported by that group for normal tissues and IDC with those we obtained, values they reported for other breast tissues are generally smaller than the ones we have obtained. Sarvazyan *et al* (1994) reported Young's modulus values of approximately 2.0 kPa and 15.0 kPa for normal tissues and IDC, respectively, which compare fairly well with the 3.25 kPa and 19.99 kPa that we obtained in this study. Given that Sarvazyan *et al* (1994) have not reported details of their measurement system, it is hard to speculate the source of the observed disagreements. Furthermore, our results do not compare well with the results obtained by Krouskop *et al* (1998). The observed disagreements in this case are attributed to the fact that, in their measurement, Krouskop *et al* (1998) applied substantial precompressional strains of 5% and 20%. Given the typical nonlinear behaviour of breast tissues illustrated in figure 4, it is reasonable to attribute part of the substantially higher values obtained by Krouskop *et al* (1998) to strain hardening.

Using our statistical analysis with a threshold p -value of 0.005, we can gain some insight into the potential of MRE for breast cancer diagnosis. These data show that fibroadenomas exhibit a statistically significant difference in Young's modulus from fat and fibroglandular tissue as well as from most breast pathologies one might encounter clinically. This is particularly relevant to the application of MRE as a means to improve the specificity of dynamic contrast enhanced MRI of the breast in which fibroadenomas can frequently enhance in a manner similar to many breast cancers. As a result, fibroadenomas present as a frequent

source of false positive findings, requiring tissue biopsy to confirm their identity. These data suggest that an accurate assessment of tissue modulus by MRE may provide a means to improve this differentiation and reduce unnecessary biopsies.

In contrast, our measurements indicate that differentiating some other tissues such as ILC, fibrocystic disease, DCIS and low- and intermediate-grade IDC on the basis of an accurate MRE measurement may not be possible. These conclusions are necessarily tempered by the fact that while this study is among the largest *in vitro* study of breast tissue biomechanical properties the individual number of tissue specimens for each tumour type was still small. Furthermore, these measurements were performed *ex vivo* and may not reflect the true behaviour of breast tissue constitutive properties *in vivo*.

These data add to the small, but critically needed, database of human biomechanical breast tissue properties which can be applied to a number of interesting research questions. Specifically, these data can guide those interested in creating biologically relevant tissue models based on finite element methods. Unlike MRE studies that probe tissues with extremely small tissue displacements and therefore are largely restricted to a linear response, these studies demonstrate clear nonlinear behaviour, which are important when applied to tissue models undergoing large tissue deformations. In such applications, tissue hyperelastic properties obtained from studies conducted by Wellman (1999) and Samani and Plewes (2004) can be used.

Furthermore, these data show clear and reproducible trends in tissue modulus consistent with existing MRE studies and empirical observations. While the issue of their relevance to the *in vivo* condition is an important question, when combined with future *in vivo* MRE studies, these data may shed insight into the changes in tissue biomechanical properties that may occur post-surgically. These may permit assessment of biomechanical factors unique to the *in vivo* state that might influence tissue biomechanical properties. An important example would be the study of the effects of alterations in interstitial tissue pressure between normal tissues and some breast tumours as reported by Nathanson and Nelson (2004), which have been identified as a possible factor in chemotherapeutic drug delivery. As such, these data when combined with *in vivo* MRE studies may elucidate interesting biological tissue properties as a result of differential biomechanical properties.

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